FastSpeech 2 for English and Telugu Speech Synthesis

Project Report — IIIT Hyderabad Research Assistant Assignment

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Base implementation: [ming024/FastSpeech2](https://github.com/ming024/FastSpeech2)

Reference paper: *FastSpeech 2: Fast and High-Quality End-to-End Text to Speech*, Ren et al., ICLR 2021

1. Project Overview

In this project, I worked on text-to-speech generation for English and Telugu using FastSpeech 2. I trained a separate model for each language and used a pre-trained HiFi-GAN vocoder to convert the predicted mel spectrograms into audio.

The work included preparing the datasets, processing text, obtaining duration labels, training the models, and testing speech generation. I also studied duration, pitch, and energy to understand how the models handled pronunciation and pauses.

The English model used multi-speaker recordings, while the Telugu model used single-speaker recordings. Each model was trained for 30,000 steps. Both models showed lower training loss over time, but validation loss stopped improving much earlier.

> About the results: The numerical values below are retained from the original experiment report. The source code, raw logs, generated audio, and plots were not included with this report. Conflicting values and details that still need verification are identified below.

2. Project Objectives

The main objectives were to:

- Adapt the FastSpeech 2 pipeline for English and Telugu.
- Convert text, numbers, dates, and times into suitable model inputs.
- Train both models with a limited amount of speech data.
- Test difficult words, context-dependent pronunciation, and unusual spaces.
- Compare training and validation losses.
- Study the duration, pitch, and energy outputs.

3. Model and Workflow

3.1 Workflow

During training, the pipeline prepares text and audio, extracts acoustic features and duration labels, and uses these targets to train FastSpeech 2.

During inference, the pipeline converts an input sentence into phonemes. FastSpeech 2 predicts their duration, pitch, and energy, then generates a mel spectrogram. HiFi-GAN converts this spectrogram into a speech waveform.

3.2 Main Components

Component	Purpose
Text processor	Cleans the sentence and converts it into phoneme tokens.
Encoder	Builds a representation of the input phonemes.
Duration predictor	Estimates how many acoustic frames each phoneme should occupy.
Pitch predictor	Estimates the pitch pattern.
Energy predictor	Estimates the energy pattern.
Length regulator	Expands the phoneme sequence using its durations.
Decoder	Predicts an 80-channel mel spectrogram.
PostNet	Refines the predicted mel spectrogram.
HiFi-GAN vocoder	Converts the mel spectrogram into audio.

The PostNet is part of the chosen reference implementation. It should not be described as a component of the original FastSpeech 2 paper.

3.3 Reported Model Settings

Setting	Value
Encoder blocks	4
Decoder blocks	6
Attention heads	2
Hidden dimension	256
FFT convolution filter size	1,024
FFT convolution kernel sizes	9 and 1
Transformer dropout	0.2
Variance predictor layers	2 convolution layers
Variance predictor filter size	256
Variance predictor kernel size	3
Variance predictor dropout	0.5
Pitch and energy bins	256 each
Mel channels	80
PostNet layers	5 convolution layers

These are the settings recorded in the draft. The saved configuration files should be used to confirm the exact settings for each training run, including whether pitch and energy were represented at the frame level or phoneme level.

4. Datasets and Preprocessing

4.1 Dataset Summary

Setting	English	Telugu
Dataset	ai4bharat/Svarah	SPRINGLab/IndicTTS_Telugu
Speech type	Indian-accented English, multi-spea	Telugu, single-speaker
Reported training duration	5.00 hours	5.00 hours
Reported training utterances	3,472	1,589
Reported validation duration	1.00 hour	1.00 hour
Reported validation utterances	670	305
TTS audio sample rate	22,050 Hz	22,050 Hz
Audio format	Mono, 16-bit PCM	Mono, 16-bit PCM
Split seed	42	42

The draft records 38 English speaker entries. Speaker identity and accent category are different labels, so these entries should not automatically be described as 38 distinct accents. The final mapping should be checked in the speaker metadata.

The Telugu training and validation sets share the same speaker because the dataset is single-speaker. Therefore, this is not a speaker-independent split. For both languages, the manifests should be checked for duplicate audio and repeated transcripts. The English speaker overlap also needs to be recorded explicitly.

4.2 Preprocessing Steps

The reported preprocessing steps were:

Prepare the audio and its matching transcript.
Convert audio to the required TTS format.
Trim leading and trailing silence and apply peak normalization.
Apply spectral noise reduction to the English recordings.
Normalize text and convert it into phoneme tokens.
Obtain alignment information for duration targets.
Extract mel spectrograms, pitch, and energy.
Prepare training and validation metadata.

Noise reduction can also change speech details. Its effect should be checked by listening to recordings before and after processing.

4.3 Acoustic Features

Feature setting	Value
FFT size	1,024
Window length	1,024 samples
Hop length	256 samples
Mel channels	80
Pitch extraction	PyWorld dio and stonemask
Energy extraction	L2 norm of the STFT magnitude

The draft reports the following normalization statistics:

Statistic	English	Telugu
Pitch mean (Hz)	176.86	208.39
Pitch standard deviation (Hz)	57.12	40.10
Energy mean	48.62	41.43
Energy standard deviation	38.83	36.25

Energy values depend on the feature extraction method and are not sound levels in decibels. Normalization statistics should be fitted on the training set and reused for validation and inference. The original report does not establish whether this restriction was followed.

5. Implementation Changes

5.1 English Text Processing

I added text processing rules for times, dates, and selected words. For example, 10:45 AM was expanded into "ten forty five a m." The romanized Telugu word andaroo was assigned an approximate English phoneme sequence for a code-switching test.

I also added rules for the word *conduct*, which has different stress patterns depending on its use:

Usage	Example	Target phonemes
Noun	The orchestra's conduct	K AA1 N D AH0 K T
Verb	The researchers will conduct	K AH0 N D AH1 K T

These rules use nearby words to select a pronunciation. They cover the selected examples but do not provide a general solution for all words with multiple pronunciations. Correct phoneme input also does not guarantee that the generated audio expresses the intended stress clearly.

The tokenizer retained word boundaries using sp pause tokens. An accidental space inside a word, such as rhyth m, could therefore affect pronunciation or introduce a pause.

5.2 Alignment

The project draft describes a custom alignment script, project/acoustic_aligner.py, using TorchAudio's MMS forced alignment pipeline as an alternative to MFA on Windows.

MMS-FA aligns normalized text tokens with audio. Its default token timestamps are not automatically phoneme boundaries. A phoneme-based FastSpeech 2 pipeline therefore needs a documented conversion from those timestamps to phoneme duration labels. This conversion should be checked in the custom script.

The aligner input must also use the sample rate required by its model. The 22,050 Hz TTS sample rate should not be assumed to be the alignment model's input rate. These requirements are described in the [TorchAudio alignment tutorial](https://docs.pytorch.org/audio/2.3.0/tutorials/forced_alignment_for_multilingual_data_tutorial.html).

5.3 Telugu Text Processing

I added a rule-based Telugu grapheme-to-phoneme converter to process Telugu characters. The implementation described in the draft handles vowels, consonants, dependent vowel signs, virama, anusvara, and visarga.

A consonant normally carries an inherent vowel. The parser changes or removes that vowel when a dependent vowel sign or virama follows it. This is needed for words with consonant clusters and doubled consonants, such as ప్రభావాన్ని and తప్పు.

The original claim of exactly 50 phonemes is not retained because the listed categories do not establish that total. The exact inventory should be taken from symbols.py.

5.4 Telugu Number and Date Processing

The Telugu normalizer expands numerical expressions before phoneme conversion.

Input	Reported expansion
2026	రెండు వేల ఇరవై ఆరు
12వ	పన్నెండవ
10:45	పది నలభై ఐదు

These examples show the intended normalization behavior. Additional tests are needed for other number formats and sentence contexts.

5.5 Environment Changes

The draft describes changes to mel-filter calls, checkpoint loading, and in-memory tensor caching for the Windows environment. The exact package versions and code changes should be recorded from the working environment. The draft's version labels are not sufficient to reproduce the setup reliably.

6. Training Setup

Setting	Reported value
GPU	NVIDIA GeForce RTX 4050 Laptop GPU
GPU memory	6 GB
Operating system	Windows 11
Optimizer	Adam
Adam beta values	0.9 and 0.98
Adam epsilon	1e-9
Learning rate schedule	Noam
Warmup steps	2,000
Batch size	16
Gradient clipping	1.0
Training steps	30,000 per model

The reported losses include mel and PostNet L1 losses, along with pitch, energy, and log-duration mean squared errors. The implementation should be checked for the weights and averaging method used to calculate the total loss.

7. Training Results

7.1 Reported Total Loss

Model	Step	Training loss	Validation loss
English	1,000	5.3243	3.7585
English	10,000	1.6151	3.5395
English	20,000	1.3871	3.6199
English	30,000	1.2725	3.6240
Telugu	1,000	3.1448	2.4815
Telugu	10,000	0.8623	2.2592
Telugu	20,000	0.7610	2.2582
Telugu	30,000	0.7132	2.2700

7.2 Final Component Losses

Metric at step 30,000	English	Telugu
Training mel loss	0.5678	0.3380
Training PostNet mel loss	0.5673	0.3374
Training pitch loss	0.0515	0.0176
Training energy loss	0.0587	0.0146
Training duration loss	0.0272	0.0056
Validation mel loss	1.0120	0.6287
Validation PostNet mel loss	1.0117	0.6286

Loss consistency: Some reported totals do not match the sum of their listed components. For example, the final Telugu components sum to 0.7132, which matches its total. The final English components sum to 1.2725, which also matches. However, the original English components at step 10,000 sum to 1.3965, while its reported total is 1.6151. Other intermediate rows also contain differences. These values should be checked against the raw logs rather than corrected by guessing.

7.3 Interpretation

Training loss decreased for both models. Validation loss improved at first and then remained nearly unchanged or increased slightly. This shows a gap between fitting the training data and performing well on validation data, and it may indicate overfitting.

Among the checkpoints listed, English had its lowest validation loss at step 10,000. Telugu had its lowest validation loss at step 20,000. The final checkpoint was therefore not the best listed checkpoint for either language.

Telugu had lower reported losses than English. Consistent single-speaker recordings may have contributed to this result. However, the two datasets differ in speakers, language, recording conditions, and feature distributions. Their losses alone cannot establish which model produces better speech.

8. Speech Generation Tests

The following sentences were used in the reported tests. They cover several text-processing challenges. The descriptions explain what each test examines; they do not claim verified listening results.

8.1 English Sentences

ID	Sentence	Purpose
E1	The antidisestablishmentarianism m	Check a long word and sentence com the parliamentary
E2a	The orchestra's conduct during the	Check noun stress in “conduct.”
E2b	The researchers will conduct an exp	Check verb stress in “conduct.” od.
E3	The rhyth m of the orchestra chang	Check the effect of a space inside a word.
E4	The meeting will start at 10:45 AM	Check dates, times, and a romanized Telugu word.

8.2 Telugu Sentences

ID	Sentence	Purpose
T1	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో	Check a technical word, long vowels, and consonant clusters.
T2a	తాడుపై అతని పట్టు చాలా బలంగా	Check “పట్టు” in the sense of grip.
T2b	ఆమె పట్టు చీరను ధరించి వేడుకకు	Check “పట్టు” in the sense of silk.
T3	పులి అడ విలో నిశ్శబ్దంగా నడిచింది	Check an unusual word break and doubled consonants.
T4	ఈ సమావేశం 2026 నవంబర్ 12వ	Check number, date, and time expansion in Telugu sentences.

The Telugu word పట్టు has different meanings in T2a and T2b, but this does not require a different phoneme sequence. Differences in duration or pitch can come from sentence context and do not prove that the model understands the word's meaning.

The original report describes clear pronunciation and natural speech for these examples. Those statements need confirmation from the saved audio and a documented listening evaluation. No listener scores, word error rates, or formal intelligibility results were provided.

9. Duration, Pitch, and Energy Analysis

9.1 Duration

With a hop length of 256 samples and a sample rate of 22,050 Hz, one mel frame represents approximately 11.61 ms.

The draft reports the following examples from the predicted durations:

Example	Reported duration	Approximate time
First vowel in noun “conduct”	10 frames	116.1 ms
First vowel in verb “conduct”	8 frames	92.9 ms
Pause in rhyth m	2 frames	23.2 ms
పట్టు in T2a	41 frames	476.0 ms
పట్టు in T2b	37 frames	429.6 ms

The reported Telugu difference is four frames, or about 46.4 ms. This is a reduction of approximately 9.8% relative to T2a. These calculations are consistent with the stated frame counts, but the counts themselves still need confirmation from the saved outputs.

9.2 Pitch and Energy

The draft describes English pitch values around 160–220 Hz and Telugu values around 190–250 Hz in selected examples. It also describes higher energy near vowels and lower energy near pauses or consonant closures.

These observations should be checked against the plots. The analysis should state whether each plot shows model predictions or measurements extracted from the generated waveform. Normalized pitch values must be converted back to Hz before they are labelled as frequencies.

Duration, pitch, and energy plots help explain model behavior, but they do not replace listening tests.

10. Limitations and Next Steps

Limitation	Next step
Validation loss stops improving while training loss fal	Compare earlier checkpoints using validation results and listening tests.
English recordings contain speaker and recording var	Review the speaker mapping, sample balance, and recording quality.
Alignment-to-phoneme conversion is not fully docun	Inspect duration labels and manually check a small set of alignments.
Rule-based pronunciation handles only selected case	Add more test cases before extending the rules.
Spaces inside words can change pronunciation.	Test input correction without changing valid word boundaries.
English phonemes only approximate Telugu code-sw	Review the pronunciation mapping with Telugu speakers.
Audio quality claims lack formal evaluation.	Conduct listening tests for clarity, pronunciation, and naturalness.
Some loss values and environment details conflict.	Regenerate tables from logs and save exact environment versions.

The first priority is to verify the data, alignments, speaker selection, and checkpoint choice. These checks will make later model improvements easier to evaluate.

11. Reproducibility and Output Files

The draft refers to the following project files and output locations:

Path	Reported purpose
project/acoustic_aligner.py	Custom alignment processing
project/English/inference.py	English speech generation
project/English/text/cleaners.py	English text normalization
project/Telugu/text/telugu_phonemes.py	Telugu phoneme conversion
project/Telugu/text/telugu_numbers.py	Telugu number and date expansion
project/Results/English/test_generated_audios/	Generated English audio
project/Results/Telugu/test_generated_audios/	Generated Telugu audio
project/Results/English/variance_adaptor_plots/	English duration, pitch, and energy plots
project/Results/Telugu/variance_adaptor_plots/	Telugu duration, pitch, and energy plots

The original execution commands depend on custom scripts and have not been verified against the repository. They should not be treated as tested setup instructions. Before sharing runnable commands, confirm the script arguments, configuration paths, checkpoint paths, and working directory.

For a reproducible release, include:

- The exact dependency versions from the working environment.
- Training and validation manifests, with post-preprocessing counts and durations.
- Speaker IDs and sample counts for the English model.
- Model, preprocessing, and training configuration files.
- Alignment settings and the method used to obtain phoneme durations.
- Raw training logs and the selected checkpoint step.
- Inference commands, speaker selection, audio samples, and analysis plots.

12. Conclusion

This project explored FastSpeech 2 for English and Telugu speech synthesis. I worked on language-specific text processing, dataset preparation, model training, and analysis of duration, pitch, and energy.

Both models showed decreasing training loss, while validation improvement slowed earlier. The main lesson is that lower training loss alone is not enough to judge speech quality. Reliable alignment, suitable speaker data, checkpoint selection, and listening tests are also needed to evaluate the final system.

References

Ren et al. [FastSpeech 2: Fast and High-Quality End-to-End Text to Speech](https://arxiv.org/abs/2006.04558), ICLR 2021.

[ming024/FastSpeech2 reference implementation](https://github.com/ming024/FastSpeech2).

[TorchAudio: Forced alignment for multilingual data](https://docs.pytorch.org/audio/2.3.0/tutorials/forced_alignment_for_multilingual_data_tutorial.html).

[AI4Bharat Svarah dataset](https://huggingface.co/datasets/ai4bharat/Svarah).

[SPRINGLab IndicTTS Telugu dataset](https://huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu).